

Probability Distributions

ECON 97: Economic Toolkit

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<https://join.iclicker.com/RDNG>

Quiz: Independence

Suppose the probability of a male birth is 0.5. In a family of 4, what is the probability of there being at least one boy?

- A. 0.75
- B. 0.5
- C. 0.0625
- D. 0.9375

Agenda

- ① Recap of material
- ② Random variables
- ③ Probability distributions
- ④ Mean & variance

Important formulas:

- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- Compliment: $P(A') = 1 - P(A)$
- Mutually exclusive: $P(A \cap B) = 0$
- Exhaustive: $P(A \cup B) = 1$
- $P(A \cup B)' = P(A' \cap B')$ and $P(A \cap B)' = P(A' \cup B')$
- $P(A \cap B') = P(A) - P(A \cap B)$
- Subset: if $A \subset B$ then $P(A \cap B) = P(A)$

Review of Concepts - Week 2

- Replacement, order matters: n^k
- No replacement
 - Order matters: ${}_nP_k = \frac{n!}{(n-k)!}$
 - Order doesn't matter: ${}_nC_k = \frac{n!}{(n-k)!k!}$
- ${}_nC_k = {}_nC_{(n-k)}$
- ${}_nC_n = 1$
- ${}_nC_1 = n$
- ${}_nC_0 = 1$

Review of Concepts - Week 3

- $P(A|B) = \frac{P(A \cap B)}{P(B)}$
- $P(B|A) = \frac{P(A \cap B)}{P(A)}$
- If A, B independent ($A \perp B$):
 - $P(A \cap B) = P(A)P(B)$
 - $P(A|B) = P(A)$ and $P(B|A) = P(B)$
 - $A' \perp B'$, $A' \perp B$, and $A \perp B'$
 - $A \perp B \perp C \implies P(A \cap B \cap C) = P(A)P(B)P(C)$

Random Variables

Think of a variable X that represents the outcome of an experiment.

The outcome space is called the **support** of X

→ E.g., if X is the result of rolling a die, then the support is:

$$S = \{1, 2, 3, 4, 5, 6\}$$

Compute the probability of each outcome using a function called the **PMF**:

$$P(X = x) = f(x)$$

Here, $f(x) = 1/6$ for all values in the support.

The probabilities must add up to 1:

$$\sum_{x \in \{1, \dots, 6\}} f(x) = 1$$

PMF (probability mass function)

$$P(X = x) = f(x)$$

$$\text{E.g., } f(3) = P(X = 3)$$

The PMF must give a probability for every possible value of X

Sum of all values of the PMF is 1 $\rightarrow \sum_x f(x) = 1$

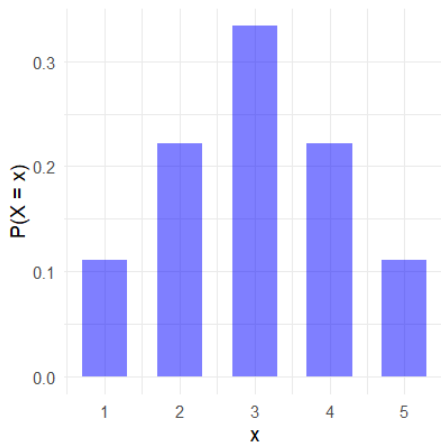
CDF (cumulative distribution function)

$$P(X \leq x) = F(x)$$

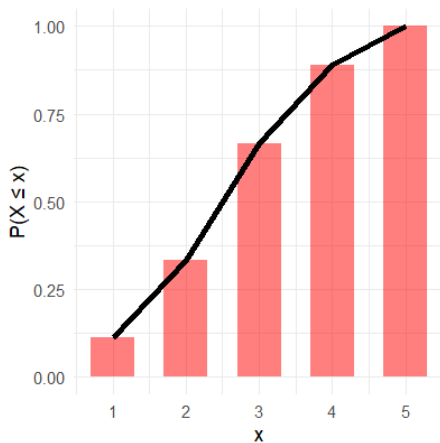
$$\text{E.g., } F(3) = P(X \leq 3) = f(0) + f(1) + f(2) + f(3)$$

Symmetric Distribution

PMF: Symmetric Distribution

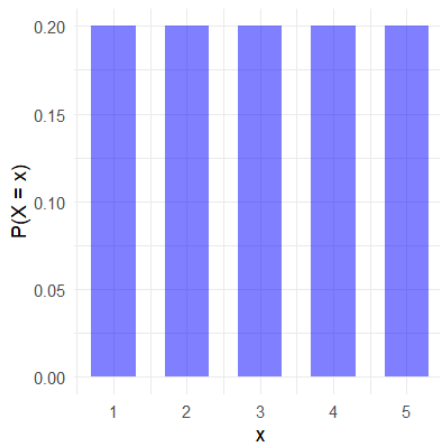


CDF: Symmetric Distribution

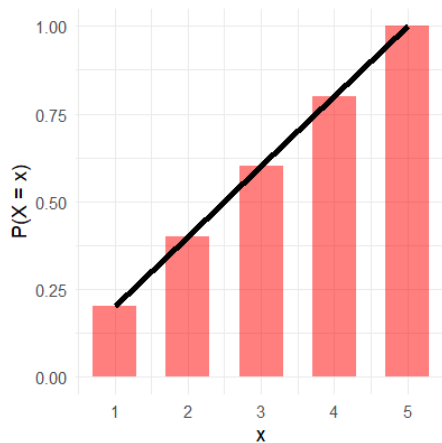


Uniform Distribution

PMF: Uniform Distribution

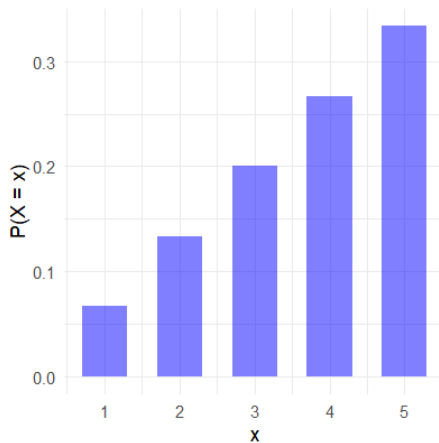


CDF: Uniform Distribution

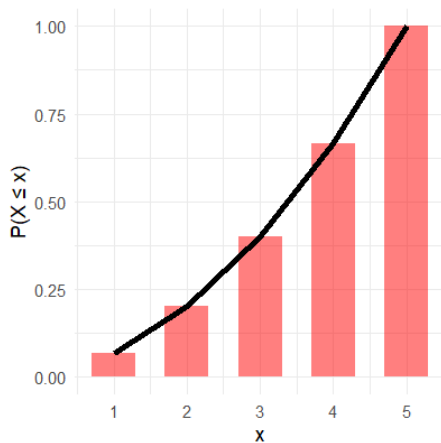


Increasing Distribution

PMF: Increasing Distribution



CDF: Increasing Distribution



Mean and Variance

Probability distributions have both a **mean** and **variance**

Mean: $\mu = \sum_{S_x} x f(x)$

Variance: $\sigma^2 = \sum_{S_x} (x - \mu)^2 f(x) = \sum_{S_x} x^2 f(x) - \mu^2$

E.g., if $f(x) = \frac{x}{6}$ with support $S_x = \{1, 2, 3\}$:

$$\begin{aligned}\mu &= f(1) * 1 + f(2) * 2 + f(3) * 3 \\ &= 1/6 + 2/6 * 2 + 3/6 * 3 \\ &= 14/6 = 7/3\end{aligned}$$

$$\begin{aligned}\sum x^2 f(x) &= f(1) * 1^2 + f(2) * 2^2 + f(3) * 3^2 \\ &= 1/6 + 2/6 * 4 + 3/6 * 9 \\ &= 36/6 = 6\end{aligned}$$

$$\sigma^2 = 6 - \left(\frac{7}{3}\right)^2 = \frac{54 - 49}{9} = \frac{5}{9}$$

Solving for the PMF

Suppose that the random variable X has probability mass function:

$$f(x) = cx^2 \text{ with support } S = \{1, 2, 3, 4\}$$

Find the following:

- The value of c , such that $f(x)$ is a valid PMF
- The mean of X (μ)
- The variance of X (σ^2)
- $P(2.5 < X \leq 4)$

Solving for the PMF